

Customer Churn Insights

Streamlit web dashboard for churn prediction, model comparison, explainability, and retention recommendations.

Customers

7,043

Churn rate

26.5%

Best model

Gradient Boosting

Best ROC-AUC

0.843

Prediction

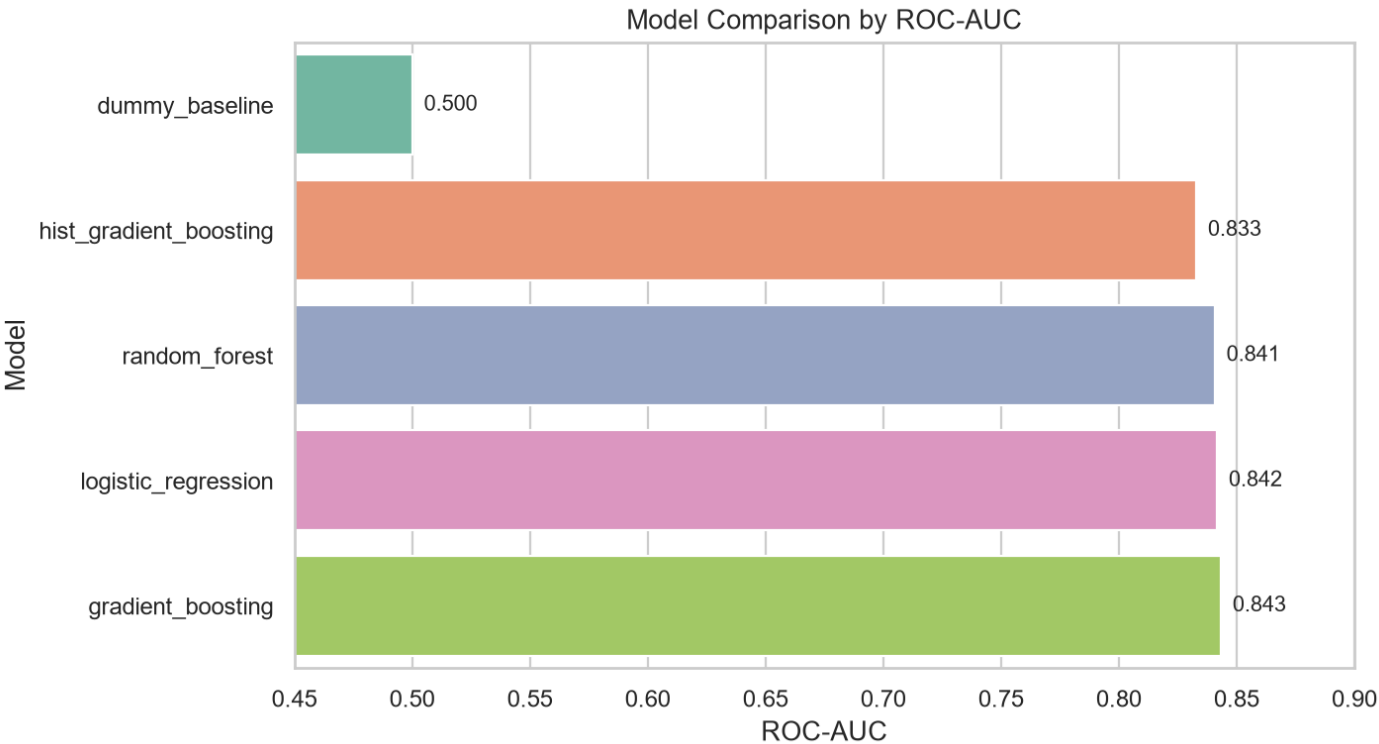
Model Performance

Threshold Tuning

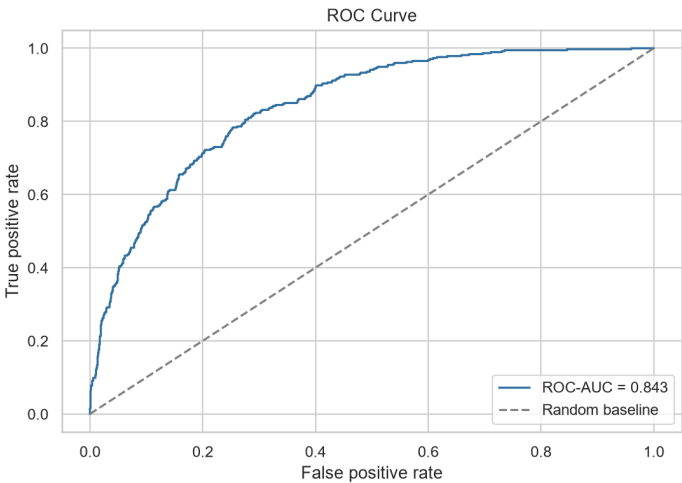
Business Insights

Model Comparison

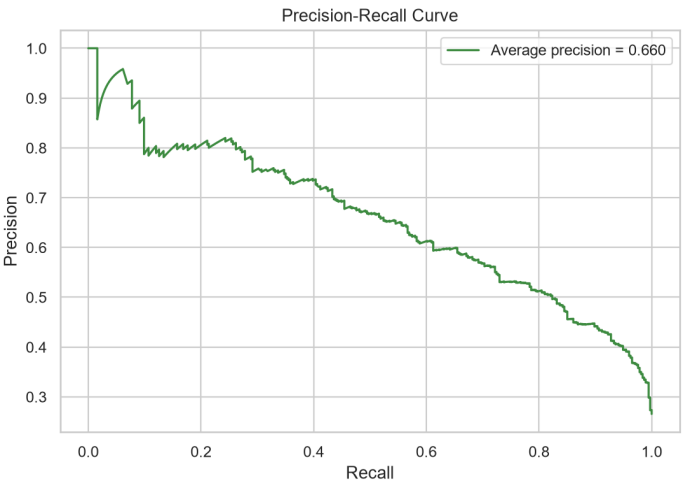
	model	accuracy	precision	recall	f1	roc_auc
0	gradient_boosting	0.803	0.666	0.516	0.581	0.843
1	logistic_regression	0.738	0.504	0.783	0.614	0.842
2	random_forest	0.764	0.539	0.775	0.636	0.841
3	hist_gradient_boosting	0.791	0.625	0.529	0.573	0.833
4	dummy_baseline	0.735	0.000	0.000	0.000	0.500



ROC-AUC comparison across trained models



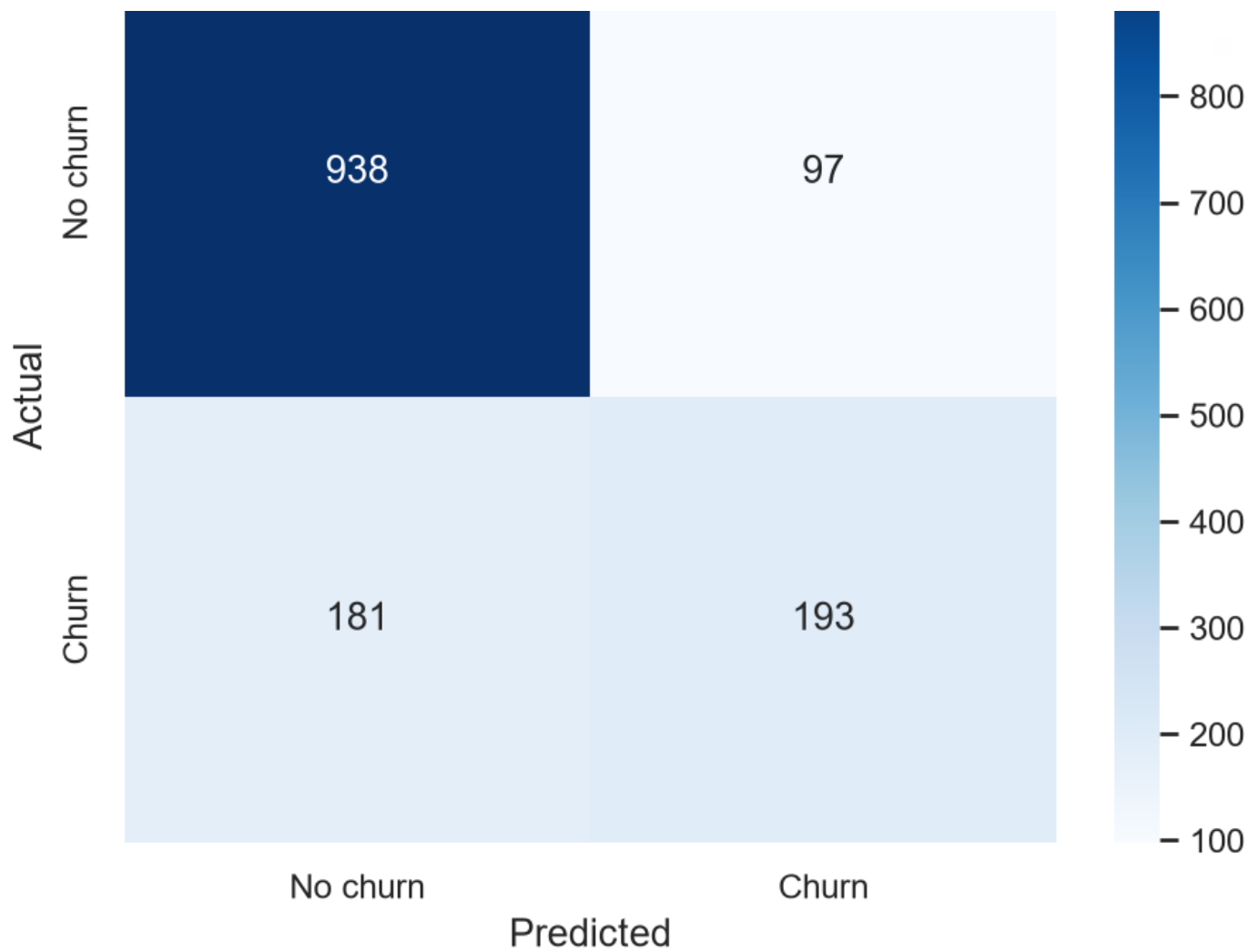
ROC curve for selected model



Precision-recall curve for selected model

Confusion Matrix at Threshold 0.50





Confusion matrix at default threshold 0.50